ADITYA UNIVERSITY

Freshman Engineering Department
Department of Mathematics
Practice Questions

AY: 2025-26 **SEM-II**

Course Name & Code: Differential Equations and Vector Calculus (2501MA02).

Unit-V: Vector Integration.

1. If $\bar{F} = 3xy\,\bar{\imath} - y^2\bar{\jmath}$, evaluate $\int_C \bar{F}\,.\,\overline{dr}$ where C is the curve $y = 2x^2$ in xy-plane from $(0,0)$ to $(1,2)$.
2. Find the work done in moving a particle in the force field $\bar{F} = 3x^2\,\bar{\imath} + (2xz - y)\bar{\jmath} + z\bar{k}$ along the straight line from $(0,0,0)$ to $(2,1,3)$.
3. Find the work done by the force field $\bar{F} = z\,\bar{\imath} + x\bar{\jmath} + y\bar{k}$, when it moves a particle along the arc of the curve $\bar{r} = \cos t\,\bar{\imath} + \sin t\,\bar{\jmath} - t\,\bar{k}$ from $t = 0$ to $t = 2\pi$.
4. Evaluate $\int_S \bar{F}\,.\,\bar{n}\,ds$ where $\bar{F} = 12x^2y\,\bar{\imath} - 3yz\bar{\jmath} + 2z\bar{k}$ and S is the portion of the plane $x + y + z = 1$ included in the first octant.
5. Evaluate $\int_S \bar{F}\,.\,\bar{n}\,ds$ where $\bar{F} = z\,\bar{\imath} + x\,\bar{\jmath} - 3y^2z\,\bar{k}$ and S is the surface $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.
6. Evaluate $\int_V \bar{F}dv$, given that $\bar{F} = 2xz\,\bar{\imath} - x\bar{\jmath} + y^2\,\bar{k}$ and V is the region bounded by the surfaces $x = 0, x = 2; y = 0, y = 6$ and $z = x^2,\ z = 4$.
7. Verify Green's theorem for $\int_C [(xy + y^2)dx + x^2\,dy]$, where C is bounded by $y = x$ and $y = x^2$.
8. Using Green's theorem, evaluate $\int_C [(y - \sin x)dx + \cos x\,dy]$, where C is the triangle enclosed by the lines $y = 0, x = \frac{\pi}{2}$ and $y = \frac{2}{\pi}x$.
9. Verify Stoke's theorem for $\bar{F} = (x^2 + y^2)\bar{\imath} - 2xy\bar{\jmath}$ taken around the rectangle bounded by the lines $x = \pm a, y = 0$ and $y = b$.
10. Verify Stoke's theorem for the vector field $\bar{F} = (2x - y)\bar{\imath} - yz^2\bar{\jmath} - y^2z\bar{k}$ over the upper half of the sphere $x^2 + y^2 + z^2 = 1$ bounded by its projection on the xy-plane.
11. Verify Gauss Divergence theorem for $\bar{F} = (x^2 - yz)\bar{\imath} + (y^2 - zx)\bar{\jmath} + (z^2 - xy)\bar{k}$ taken over the rectangular parallelepiped $0 \le x \le a, 0 \le y \le b,\ 0 \le z \le c$.
12. Evaluate $\iint_S (x\,dy\,dz + y\,dz\,dx + zdx\,dy)$ by using divergence theorem , where S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$.
13. State the following vector integral theorems:
    (i). Green's Theorem.
    (ii). Stokes Theorem.
    (iii). Gauss Divergence Theorem.

****